FedFusionQuant (FFQ): Federated Learning With Feature Fusion and Model Quantisation for Human Activity Recognition Using CSI

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***Abstract*—Human Activity Recognition (HAR) using Channel State Information (CSI) enables energy-efficient and non-invasive healthcare monitoring. However, conventional HAR systems rely on centralised model training, which requires the sharing of raw data, leading to privacy risks, excessive bandwidth usage, and high communication latency that limit scalability. This paper proposes FedFusionQuant (FFQ), a federated learning (FL) framework that jointly performs feature fusion, adaptive aggregation, and quantisation-aware compression during training. A novel federated distance (FedDist) mechanism dynamically adjusts parameter updates using neuron dissimilarity metrics, enhancing generalisation across heterogeneous clients. Meanwhile, quantisation-aware training (QAT) reduces model size and transmission cost while preserving accuracy. Extensive experiments on real CSI data from 30 participants demonstrate that FFQ improves multi-class HAR accuracy by 4.29% and binary fall detection by 5.55% compared to raw fusion models. Furthermore, model compression with QAT achieves a 47% reduction in communication overhead while maintaining accuracy comparable to state-of-the-art methods.**

***Index Terms*—Human activity recognition, channel state information, federated learning, model compression, data privacy, and feature fusion.**

I. Introduction

THE advent of pervasive computing has significantly transformed indoor living spaces by integrating smart devices and intelligent sensors. This technological synergy facilitates non-intrusive services like real-time monitoring, security surveillance, and automation of building and industrial processes, enabling scalable local autonomy [1], [2]. Human activity recognition (HAR) and indoor localisation are at the forefront of this technological evolution, which has emerged as a pivotal research domain. Their applicability spans diverse areas, including smart healthcare, intelligent building control, automation, the Internet of Things (IoT), behaviour analysis, gesture recognition, and smart surveillance systems [3], [4]. A key application of HAR systems lies in elderly care, particularly for fall detection. These systems empower hospital and care-home staff to continuously monitor the activities of the elderly using advanced communication and sensing technologies [5].

HAR is a classification problem that aims to identify individuals' actions and movements based on observations captured through various sensors. Recently, many studies have explored HAR using wearable sensors, ambient devices, vision sensors, and smartphones. HAR systems can broadly be categorised into vision-based and sensor-based approaches [6]. Vision-based HAR, while effective, faces significant challenges, such as privacy and sensitivity to external factors like lighting conditions, user background, and camera resolution, which underscore the need for alternative methods [7].

Transitioning from vision-based approaches, sensor-based HAR offers alternative modalities, spanning wearable and non-invasive sensing. Wearable devices such as pedometers, accelerometers, and gyroscopes require continuous use, which can be inconvenient or uncomfortable in contexts like elderly care [5], [8]. Smartphones and smartwatches, although common for activity monitoring, are not fully contactless since they must be carried, raising concerns about usability and continuous engagement [6], [8].

In contrast, radio frequency (RF) sensing has emerged as a promising non-invasive and privacy-preserving solution. Leveraging channel state information (CSI) or received signal strength indicator (RSSI), RF-based systems can effectively monitor indoor activities. CSI-based HAR is particularly advantageous, capturing fine-grained sub-carrier information that reflects human-induced scattering patterns and enables accurate activity classification [4], [6]. However, system performance depends heavily on data quality and volume, which are critical for achieving robust model accuracy and reliability [7]. Despite these benefits, current non-invasive HAR systems face key challenges due to the inherent unpredictability of wireless channels, especially multipath effects and environmental variations, that introduce signal noise and ambiguity [9].

Additionally, most existing HAR systems rely on centralised data collection and processing. This approach necessitates

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sharing raw data, raising significant privacy concerns, especially under stringent regulations like the General Data Protection Regulation (GDPR) [7], [8]. Centralised processing also leads to inefficient bandwidth usage, as large volumes of raw data must be transmitted to a central server, increasing network and storage costs. Furthermore, this setup is energy-inefficient and introduces latency, limiting the real-time performance and scalability of HAR applications. Therefore, to overcome these challenges, McMahan et al. proposed a distributed learning technique termed federated learning (FL), which offers the benefits of collaborative model training without sharing raw user data [10]. This approach offers privacy by design, solves latency issues, and reduces the communication overhead incurred by raw data sharing.

Despite the promising advantages of FL in preserving privacy and reducing latency, its application to non-invasive HAR remains limited. Existing studies often overlook key challenges such as data heterogeneity, energy efficiency, and the communication overhead associated with the training process [11]. The diversity of data collected across users and environments poses substantial difficulties in achieving consistent and accurate model performance. Furthermore, frequent exchange of model parameters between devices and the central server incurs significant communication cost, hindering the scalability of FL frameworks for HAR [10], [11]. As most sensors are battery-powered, transmitting large model updates further depletes energy resources. Addressing these challenges is therefore vital to realise the full potential of FL in enhancing the efficiency, accuracy, and privacy of HAR systems.

Therefore, addressing the critical challenges of data heterogeneity, communication overhead, and energy efficiency in HAR, this study introduces FedFusionQuant (FFQ), a FL framework that utilises signal processing, feature fusion, and model compression. Feature fusion combines statistical and differential features with processed CSI data, providing the model with the contextual understanding necessary for navigating varied environmental conditions [4], [6]. Distinguishing itself from traditional FL approaches, FFQ adopts the federated distance (FedDist) algorithm [12], tailored to mitigate overfitting by adjusting the model parameters based on a dissimilarity measure of diverging neurons, a pivotal enhancement for handling heterogeneous data effectively.

To reduce communication overhead during federated training, the proposed framework employs model compression by quantising parameters from 32-bit floating point to 16- and 8-bit integer precision. Quantisation-aware training (QAT) is also adopted, adjusting the bit precision of computations and gradients during local training to improve communication efficiency. This approach not only reduces the size of model updates exchanged in each communication round but also enables local models to adapt to reduced precision, maintaining accuracy while lowering bandwidth requirements. Furthermore, FFQ optimises the aggregation of model updates, ensuring that the collective intelligence accurately captures diverse human activities across heterogeneous environments, thereby enhancing the overall accuracy and reliability of the HAR system. The main contributions of this work are summarised as follows:

- This work introduces the FFQ framework, merging FL with edge-based preprocessing and feature fusion to enrich HAR models with a deep contextual understanding of CSI data. By extracting and fusing the statistical and differential features, FFQ aims to achieve a low false positive rate across diverse environments, particularly for fall detection.
- The study adopted the FedDist algorithm for global model aggregation, tailored to address the challenge of overfitting in the context of highly heterogeneous data. This study also modified the divergence metric by adjusting model parameters based on a dissimilarity measure of diverging neurons, resulting in the effective handling of diverse data. The results of the modified FedDist are compared with FedAVG algorithm.
- The proposed FFQ framework incorporates QAT into its model compression strategy, significantly improving the efficiency of FL architectures by training local models directly in a reduced 16 and 8-bit precision format. This approach minimises the size of model updates, which significantly reduces communication overhead while maintaining comparable accuracy with the state-of-the-art techniques.

The rest of the paper is organised as follows: Section II covers the related works, whereas Section III presents the system model, and Section IV discusses the proposed FFQ model for HAR. Section V discusses the simulation setup, and the results are discussed in Section VI. Finally, Section VII presents a Conclusion and future research directions.

II. Related Work

This section reviews the evolution of HAR, with particular focus on the integration of FL and feature fusion within HAR systems. It highlights key advancements and limitations, emphasising the transition from RSSI- to CSI-based approaches. Furthermore, it discusses the emerging convergence of FL and HAR, outlining the opportunities and challenges at their intersection.

A. RSSI-Based HAR System

As discussed in the introduction, RF sensing HAR is popular because it uses existing indoor Wi-Fi signals and requires no special equipment for data collection. The common contactless techniques for indoor HAR are RSSI and CSI, which utilise the broadcast nature of wireless signals to create unique signatures for each activity [3], [5]. RSSI-based systems use Wi-Fi fingerprinting for localisation, activity, and gesture recognition. The combination of advanced signal processing and machine learning (ML) has demonstrated the use of RSSI in gesture recognition and indoor HAR.

For instance, the authors in [13] proposed an RSSI-based HAR system to classify daily activities such as sitting, standing, and walking, achieving accuracies between 75% and 92% using a k-nearest neighbour (k-NN) classifier. In [14], a three-layer gesture recognition framework, Wi-Gest, was introduced, employing RSSI to recognise hand gestures. Tested with one and three access points, the system achieved accuracies of 87.5% and 96%, respectively. Although RSSI-based HAR is simple

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and effective, it faces challenges including multipath fading, signal distortion, device constraints, and interference from indoor objects [14], [15]. Moreover, its low granularity limits classification accuracy, while the absence of frequency diversity, providing only a single value per packet, reduces coverage and makes RSSI measurements more vulnerable to environmental variations and multipath effects [13].

B. CSI-Based HAR System

To address the inherent issues in RSSI-based systems, the focus shifted towards exploiting CSI, known for its granularity in capturing physical layer signal information. The CSI-based systems capture the fine-grained information, which includes amplitude and phase per sub-carrier of orthogonal frequency division multiplexing (OFDM) [9]. This information facilitates a deeper understanding of the environmental dynamics influenced by human activities, making it particularly suited for complex HAR scenarios, including indoor HAR, fall detection, and vital signs monitoring [3], [16]. A detailed theoretical study on non-invasive, contactless sensing using Wi-Fi CSI is presented in [17]. This study presented a fresnel zone model using contactless RF for vital sign classification and macro-level activity recognition. Macro-level activities cause reflection and attenuation in signal strength, which can be distinguished using advanced signal processing.

Similarly, the authors in [5] developed a device-free HAR and fall detection system using fine-grained CSI, classifying daily activities such as sitting, running, walking, standing, and falling via support vector machine (SVM) and long short-term memory (LSTM) networks, achieving an accuracy of approximately 95%. In [18], a CSI-based activity recognition and monitoring system (CARM) was presented, exploiting Wi-Fi signal dynamics for indoor activity monitoring. A CSI-derived speed model quantified human motion speed and signal strength variation, which was then used to construct an activity classification model linking movement speed to activity type. Singular value decomposition (SVD) was applied for data pre-processing, while random forest achieved an accuracy of around 96.

In [19], a super learner classification algorithm is proposed for indoor localisation and activity monitoring using CSI. Rigorous signal processing filters the raw data, and results are compared with the classical ML models like SVM, RF and extra tree. Similarly, one of our previous studies proposed a deep-gated recurrent unit (DGRU) based model for non-obtrusive HAR using CSI [20]. The proposed scheme used empirical model decomposition (EMD) for de-noising the raw data samples and discrete wavelet transforms for feature extraction, whereas linear discriminant analysis (LDA) for dimensionality reduction, respectively. The results are compared with LSTM and RF, and the proposed scheme achieved an accuracy of 98.12%, compared to 87% and 82%, respectively.

C. FL-Based HAR System

The increasing computational capabilities of mobile devices and the growing concerns over data privacy regulations have propelled the adoption of FL in HAR. FL, as a decentralised ML approach, allows for the aggregation of model updates without needing direct access to the raw data. This technique ensures user privacy and leverages decentralised data, offering a robust framework for HAR in ubiquitous computing environments. To this extent, a limited number of studies have been conducted on HAR leveraging FL. For instance, the authors in [11], propose a novel method for HAR using wireless signals that addresses data privacy and the non-IID nature of distributed datasets. This scheme introduced a cross-domain federated learning framework (CDFL) that leverages transfer learning and achieves high accuracy rates in ultrasonic signal-based gesture recognition tasks.

Similarly, the authors in [25] propose FedHAR as a customised FL framework that addresses significant challenges in HAR, such as label scarcity, real-time processing, and heterogeneous data. It leverages distributed learning to keep data local and employs a semi-supervised online learning strategy to handle unlabeled data effectively. This framework incorporates a hierarchical attention architecture for feature alignment and a semi-supervised learning loss to integrate gradients from labelled and unlabelled clients. After extensive testing on two public datasets, FedHAR has shown superior performance compared to existing methods, with a significant improvement of about 10% across various metrics when fine-tuning models for unlabeled clients, highlighting its effectiveness in real-world HAR applications.

The authors in [21] proposed a novel HAR method using a 3D convolutional neural network (3D-CNN) applied to sensory data, translating temporal and frequency bio-signals into voxel intensities. The approach enhances privacy and security through bitwise XOR encryption integrated with FL, achieving high accuracy with minimal degradation when data are encrypted. Similarly, [22] introduced Hydra, a hybrid-model FL framework for devices with varying computational capabilities. Using BranchyNet technology, Hydra partitions model layers across heterogeneous devices and employs clustering based on data similarity to improve co-training efficiency. It also adopts a sample selection algorithm and a large-to-small knowledge distillation technique to enhance accuracy. Extensive experiments confirmed Hydra's superior performance compared with state-of-the-art methods. However, both [21], [22] relied on wearable sensing data and overlooked communication overhead and energy efficiency. In [23], a cluster-based FL algorithm was proposed for activity classification using wearable devices, demonstrating improved model generalisation and overall system performance across diverse datasets.

It is also worth noting that, outside the HAR domain, recent cellular-network studies have explored federated edge learning (FEEL) for physical-layer CSI-related tasks. For instance, Cui et al. [26] provide an overview of FEEL for the wireless physical layer, highlighting its role in CSI acquisition, beamforming, and feedback optimisation. Similarly, Cui et al. [27] propose a communication-efficient personalised FEEL framework for massive MIMO CSI feedback, using quantised model updates and per-user decoder adaptation to handle heterogeneity. Although these works share the goal of reducing communication overhead in distributed learning, their focus remains on cellular

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TABLE I
COMPARATIVE ANALYSIS OF RECENT ADVANCES IN HAR EMPHASISING FL AND CSI

Ref	Feature Fusion	FL	Model Compression	Contributions	Limitations
[13]	✗	✗	✗	Introduced a simple RSSI-based HAR using K-means for activity classification.	Suffers from low accuracy and lacks privacy-preserving mechanisms.
[14]	✗	✗	✗	Developed a gesture recognition system emphasising simplicity and ease of use.	Relies on centralised data processing, raising privacy concerns.
[3]	✗	✗	✗	Applied ML algorithms for vital sign and activity classification using CSI data.	Centralised model poses privacy risks; does not leverage FL for data protection.
[17], [18]	✓	✗	✗	Feature-level fusion in CSI-based activity classification for improved accuracy.	Limited by a theoretical approach and lacks practical privacy solutions.
[11]	✗	✓	✗	Cross-domain FL to address HAR, considering the impact on communication overhead.	Did not incorporate feature fusion, missing out on richer contextual data analysis.
[21]	✗	✓	✗	Translated temporal and frequency bio- signal values into voxel intensities and also use encryption to ensure security.	Wearable sensor data and did not consider energy efficiency and communication overhead.
[22], [23, [24]	✗	✓	✗	Hybrid models allow co-training with varying computational capabilities.	Wearable sensing and did not consider energy efficiency and communication overhead.
Our Work	✓	✓	✓	Feature fusion, model compression with QAT to reduce communication overhead and improve energy efficiency.	-

physical-layer signal reconstruction rather than high-level human activity inference. In contrast, our study advances WLAN CSI-based HAR, where the objectives, data dynamics, and design constraints differ fundamentally. The proposed FFQ framework uniquely integrates temporal feature fusion, QAT, and FedDist-based selective aggregation to achieve communication-efficient and robust federated learning under heterogeneous non-IID conditions.

D. Literature Survey Summary

Finally, Table I summarises the contributions and limitations of the related work. Although FL presents a promising solution to privacy and scalability challenges in HAR, the integration with CSI-based systems remains underexplored. Moreover, the potential of utilising raw signal data to extract features and provide contextual information in HAR systems, particularly in an FL framework, has yet to be fully realised. However, communication overhead remains a critical issue, impacting scalability and efficiency.

Developing and implementing efficient model-sharing techniques within FL-based HAR systems could significantly mitigate this challenge, paving the way for more sophisticated, real-time activity recognition solutions. Furthermore, the majority of sensors are expected to be battery-operated, and frequently sharing significant model parameters during the training process will drain the battery. Therefore, energy efficiency in the model training process is also very crucial.

III. PRELIMINARIES AND SYSTEM MODEL

This section presents the generic FL training process for a typical classification problem for HAR. The conventional HAR model collects the raw data samples and transmits them to a centralised location for processing and model training. In contrast, the FL approach trains the model across many clients where the dataset is highly decentralised. This work considered a cross-silo FL architecture where the data is divided among a small number of trusted end clients, as shown in Fig. 1. Consider a FL system with N edge nodes (EN) representing clients, where each client i holds a local dataset D_i. The global learning objective is to minimise the loss function $F(\omega)$ over the global model parameters ω, formulated as:

$$F(\omega) = \frac{1}{N} \sum_{i=1}^{N} F_i(\omega), \tag{1}$$

where $F_i(\omega)$ is the local loss function computed on the i-th client's dataset D_i. The local loss function typically represents the average loss over the data points in D_i, expressed as:

$$F_i(\omega) = \frac{1}{|D_i|} \sum_{j \in D_i} \ell(x_j, y_j; \omega), \tag{2}$$

here, $\ell(x_j, y_j; \omega)$ denotes the loss of the model parameterised by ω on a data point (x_j, y_j) from client i's dataset, with x_j being the feature vector and y_j the associated class label.

The FedAVG algorithm aggregates the model parameters on the federated server (FS) during the training process across T communication rounds, each comprising E local epochs of training on each client, with a local batch size of B. The FedAVG process is mathematically represented through the following iterative process:

- *Initialisation:* The FS initialises the global model parameters ω^0 and distributes them to all EN. This process is represented by step ① as shown in Fig. 1.
- *Local Training:* For each communication round $t = 1, 2, \ldots, T$, and each client i, the client updates the model parameters ω by performing E epochs of gradient descent on its local dataset D_i, using a batch size of B. This can be mathematically depicted as:

$$\omega_i^{t+1} = \omega_i^t - \eta \nabla F_i(\omega_i^t), \tag{3}$$

where η is the learning rate and $\nabla F_i(\omega_i^t)$ is the gradient of the loss function F_i with respect to the model parameters ω at round t. The process of local learning is represented by step ②, as shown in Fig. 1.

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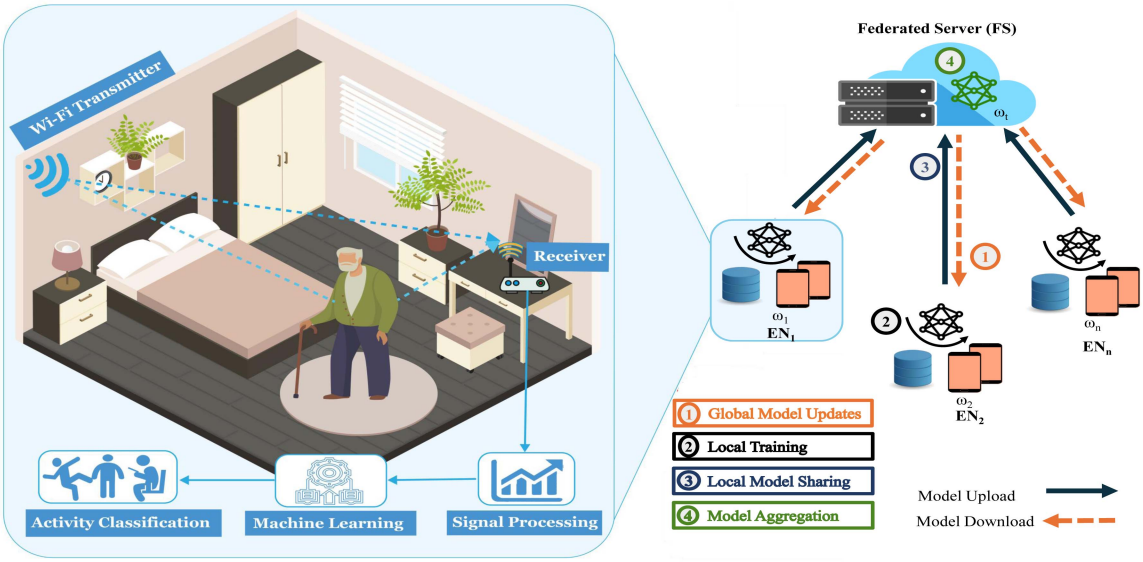


Fig. 1. The conceptual framework for FL-based HAR using CSI.

- *Model Aggregation:* After local training, each client sends its updated model parameters ω_i^{t+1} back to the FS given as step ③. The server then aggregates these parameters to update the global model step ④ as shown in Fig 1. The aggregation is performed as a weighted average, considering the size of each client's dataset:

$$\omega^{t+1} = \frac{1}{\sum_{i=1}^{N} |D_i|} \sum_{i=1}^{N} |D_i| \omega_i^{t+1} \tag{4}$$

- *Global Model Update:* The updated global model parameters ω^{t+1} are then shared with all clients, marking the beginning of the next training round.

This iterative process of local training and global aggregation continues for T rounds, aiming to minimise the global loss function $F(\omega)$ and improve the model's performance on the multi-class classification task.

IV. Proposed FFQ for HAR

This section details the proposed FFQ framework for HAR using CSI, building on the foundational understanding of FL discussed in Section III. The framework includes data preprocessing, feature engineering, feature fusion, and model compression to enhance communication efficiency and maintain high classification accuracy. We will cover our methodologies in data preprocessing, feature engineering, and innovative temporal feature extraction and fusion. Additionally, it explains the local training processes, QAT-based model quantisation, and our custom model aggregation mechanism, which collectively enhance the robustness and efficacy of the FFQ framework.

A. Signal Propagation and CSI Acquisition

The first step in FL-based HAR is to collect fine-grained CSI data under the influence of different daily routine activities. Wi-Fi signals undergo various effects due to their broadcast

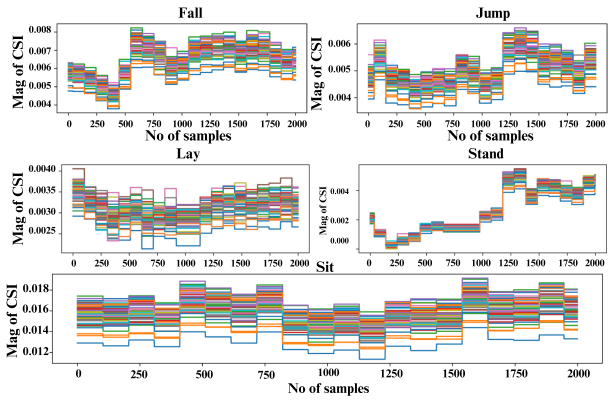


Fig. 2. 2D plots of received CSI under the influence of different activities.

nature, such as reflection, refraction, scattering, and diffraction [20]. In this study, we implemented the IEEE 802.11a standard using software-defined radios (SDR) and used two universal software radio peripherals (USRP) as transmitters and receivers. The preamble (known data sequences) in the message packet is used for channel estimation, and the physical layer CSI is acquired under the influence of different activities. The details of implementing the IEEE 802.11a standard and the data collection process are beyond the scope of this work. However, the implementation and process of raw data collection in a lab environment are discussed in our previous studies [20], [28]. The plots of the 2000 CSI samples under the influence of different activities are shown in Fig. 2. It can be observed that different activities exhibit significant variations in received CSI, which can be translated into unique signatures for each activity after careful signal processing.

B. Data Pre-Processing and Feature Engineering

The collected raw CSI signal contains information on both phase and amplitude. The received signal phase is usually affected by carrier frequency offset (CFO) caused by a mismatch

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of transmitter-receiver clocks. A small CFO causes significant variation in the phase of the received signal, which makes the phase changes caused by the body negligible [4]. Therefore, this work considered only the amplitude information. Although the amplitude of the received signal is considered a stable metric, the noise added by the external environment is still a big problem. The noisy CSI may not give distinct features; therefore, data denoising is one of the vital steps in HAR.

1) Signal Denoising Using EMD: In this work, we used EMD for denoising, given the non-linear and non-stationary nature of time series CSI data [4], [17]. The EMD decomposes the raw symbols, which consists of m samples for each client i, denoted as $X(t) = \{x_1, x_2, \ldots, x_m\}$. The EMD process enables the decomposition of $X(t)$ into a finite set of intrinsic mode functions (IMFs) and a residue, significantly enhancing the signal quality for further processing. For client i, the denoising process through EMD can be mathematically represented as:

$$X_i(t) = \sum_{k=1}^{n_i} IMF_{ik}(t) + r_i(t), \tag{5}$$

where $IMF_{ik}(t)$ are the intrinsic mode functions extracted from $X_i(t)$, n_i denotes the total number of IMFs for client i, and $r_i(t)$ represents the residue. The signal post-denoising, denoted as $\hat{X}_i(t)$, is reconstructed by excluding the first few $IMF_{ik}(t)$, identified primarily as noise components:

$$\hat{X}_i(t) = \sum_{k=p+1}^{n_i} IMF_{ik}(t) + r_i(t), \tag{6}$$

where p is the last IMF considered to be noise. The practice of identifying the first few $IMF_{ik}(t)$ as noise components is grounded by the characteristic of EMD decomposition, where higher-frequency IMFs primarily capture environmental noise and transient interference in CSI signals, which typically have minimal correlation with human activity-induced fluctuations [29], [30]. These noise components are distinct from the lower-frequency, structured patterns of human movement that reside in subsequent IMFs. The critical parameter p, denoting the last noise-dominant IMF to be excluded, was determined empirically. We conducted a systematic search across a validation dataset, evaluating activity recognition performance for varying p values. The optimal p was selected to maximise classification accuracy, ensuring effective noise removal while preserving activity-relevant information. For our specific CSI dataset, we found that setting $p = 2$ provided the best balance, aligning with common practices in EMD-based sensor data denoising where the first few modes often represent noise [31].

2) Feature Engineering: Feature engineering plays a crucial role in the subsequent analysis following the denoising of the time series data. Given its efficacy in capturing both frequency and time domain aspects of human activities, we employed the discrete wavelet transform (DWT) for feature extraction. The denoised signal, denoted as $\hat{X}_i(t)$, undergoes decomposition into high and low-frequency components using DWT, facilitating the separation of the signal into different time scales. This process involves passing $\hat{X}_i(t)$ through a pair of low and high-pass filters, followed by downsampling to adhere to the Nyquist criteria. The extraction of meaningful features is achieved by applying a thresholding technique to the wavelet coefficients obtained from DWT, which is mathematically represented as [32]:

$$\tilde{\Phi}_{j,i} = \begin{cases} (\Phi_{j,i})\left(|\Phi_{j,i}| - \tau_j\right), & |\Phi_{j,i}| \geq \tau_j \\ 0, & |\Phi_{j,i}| < \tau_j \end{cases} \tag{7}$$

where $\Phi_{j,i}$ and $\tilde{\Phi}_{j,i}$ denote the original and denoised wavelet coefficients, respectively, at the jth level of decomposition and ith location. The threshold τ_j is determined based on the universal threshold τ_U, detailed as follows [33]:

$$\tau_U = \sigma\sqrt{2\log(L)}, \tag{8}$$

with L representing the length of the pre-processed CSI signal, and σ denoting the standard deviation of noise, estimated by:

$$\sigma = \frac{\text{median}(|Z_1|)}{0.6745}, \tag{9}$$

where Z_1 is the first level detail coefficient. After completing the feature extraction process, the next step is to perform feature selection to reduce redundancy in the training data and avoid overfitting. While there are many ways to select features, we chose LDA because the CSI data shows a normal distribution with features that are minimally correlated [34]. LDA aims to create a new subspace of the data by maximising the distance between classes (inter-class) and minimising the distance within the same class (intra-class), as defined by the equations below:

$$C_b = \sum_{j=1}^{k} d_j (m_j - c)(m_j - c)^T, \tag{10}$$

where d_j denotes the number of samples in the jth class, m_j the mean vector of the jth class, and c the global centroid. The within-class scatter matrix C_w is defined as:

$$C_w = \sum_{j=1}^{k} \sum_{x \in D_j} (x - m_j)(x - m_j)^T, \tag{11}$$

where D_j represents the set of all samples in the jth class. Furthermore, the total scatter matrix C_t is represented by:

$$C_t = \sum_{j=1}^{n} (x_j - c)(x_j - c)^T, \tag{12}$$

where x_j are the data points, and c is the global centroid. The lower dimensions of the input training data are achieved through the application of eigenvalue decomposition on the scatter matrices, which effectively eliminates noisy and redundant components.

C. Temporal Feature Extraction and Feature Fusion

This study extracts temporal features from the denoised signal $\hat{X}_i(t)$ to expand the feature space beyond the conventional LDA feature set. The goal is to capture the dynamic nature of human activities with statistical and first-order differential features from each signal window. A comprehensive set of statistical features, including mean, variance, standard deviation, skewness, and kurtosis, is computed from each window W in the denoised signal $\hat{X}_i(t)$ to capture the signal's intrinsic properties over time.

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First-order differential features are calculated to encapsulate the temporal evolution of the signal, highlighting changes between consecutive samples. For a signal window W, the differential feature, $Diff_i^W$, is defined as:

$$Diff_i^W = \{x_{i+1} - x_i | x_i, x_{i+1} \in W\} \tag{13}$$

After extracting the statistical features (S_i^W) and differential features ($Diff_i^W$), a fusion process is performed to combine these temporal features with the raw CSI features selected through LDA (G_i'). This approach ensures a more complete characterisation of each activity, considering both the static and dynamic aspects of the signal. The fusion of LDA-selected features and temporal features yields the final feature vector H_i for each signal window W, formulated as:

$$H_i = G_i' \oplus S_i^W \oplus Diff_i^W \tag{14}$$

In this formulation, $\oplus$ symbolises the concatenation operation, blending the diverse feature sets into a unified representation H_i also named as combined temporal features (CTF), which is the input for the subsequent classification model. This fused feature vector H_i is poised to offer a robust foundation for accurate activity recognition, embodying the multifaceted nature of human motion patterns.

D. Local Training and Quantisation

A pivotal stage in our proposed FFQ framework is local training, wherein individual clients fine-tune the global model using their local data D_i as discussed in Section III. This stage is crucial for adapting the global model to diverse local conditions and data characteristics. Following local training, the quantisation process optimises communication efficiency by compressing the updated model parameters before transmitting them back to the FS.

1) Quantisation Process: The quantisation strategy aims to reduce the model size, facilitating efficient communication across the FL network. This reduction is achieved by lowering the precision of the model parameters to a specified bit-length b without significantly compromising the model's predictive performance. The simplified quantisation operation is represented mathematically as:

$$\hat{\omega}_i^{t+1} = Q(\omega_i^{t+1}, b), \tag{15}$$

where $Q(\cdot, b)$ denotes the quantisation function that compresses the parameters ω_i^{t+1}, with b indicating the target bit precision. This process balances the trade-off between model accuracy and communication cost.

Formally, the quantisation function is defined as:

$$Q(\omega, b) = \text{round}\left(\frac{\omega - \omega_{min}}{S}\right) \times S + \omega_{min}, \tag{16}$$

where ω_{min} and ω_{max} represent the range of model parameters, and S is the quantisation step size for the given bit-width b is given by $\frac{\omega_{max} - \omega_{min}}{2^b - 1}$.

2) Pre-Quantisation and Post-Quantisation: The quantisation process can be performed into two stages: pre-quantisation and post-quantisation. In **Pre-Quantisation**, the model parameters are compressed before local training. This phase involves standard training operations where the parameters ω_i^t are adjusted based on the local data characteristics. The goal during this phase is to adapt the global model to local nuances and improve its performance across diverse data distributions.

Post-Quantisation: After the local training, the updated model parameters ω_i^{t+1} are quantised to reduce their size before model sharing. This step compresses the parameter values, resulting in $\hat{\omega}_i^{t+1}$, which are then sent to the FS. While this significantly reduces communication overhead, it introduces potential precision loss, which can degrade the model's performance. Post-training quantisation is particularly vulnerable to such performance loss, as it is applied after training without incorporating quantisation effects into the model updates. In contrast, QAT enables the model to anticipate and adapt to quantisation noise during training, leading to superior performance under low-bit quantisation.

3) Incorporating Quantisation-Aware Training: Quantisation-Aware Training (QAT) is integrated into the local training phase to further enhance the model's robustness to quantisation effects. QAT anticipates the impact of quantisation by incorporating a simulation of the quantisation process within the training algorithm. This approach allows the model to adapt to the reduced precision and maintain high accuracy prior to post-quantisation. The mathematical representation of QAT is as follows:

Model Quantisation during Training: The model parameters undergo a simulation of quantisation and subsequent dequantisation within each training iteration, represented by:

$$\omega_{i,sim}^t = DQ(Q(\omega_i^t, b)), \tag{17}$$

where $DQ(\cdot)$ represents the de-quantisation function, reversing the quantisation effect for training purposes.

Parameter Update with Model Quantisation: This simulated environment enables the gradient descent optimisation to account for quantisation effects, mathematically expressed as:

$$\hat{\omega}_i^{t+1} = \omega_i^t - \eta \nabla F_i(\omega_{i,sim}^t), \tag{18}$$

where η is the learning rate, and ∇F_i denotes the gradient of the loss function with respect to the simulated quantised parameters. This equation is a modified version of the local update with QAT given in (3). Incorporating QAT into local training effectively counters potential accuracy declines due to reduced parameter precision. This can be reformulated to show the explicit dependence on the quantised parameters:

$$\hat{\omega}_i^{t+1} = \omega_i^t - \eta \nabla F_i \left(DQ(Q(\omega_i^t, b))\right) \tag{19}$$

This expression makes clear that gradients are computed with respect to quantised-and-dequantised parameters, allowing the model to learn robust representations that compensate for quantisation-induced distortions. In effect, QAT bridges the gap between training and deployment environments by exposing the model to quantisation noise during learning.

4) Convergence Analysis Under QAT: We analyse the convergence behaviour of the proposed framework under QAT in a FL setting. Let each client's local objective function $F_i(\omega)$ be

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L-smooth and μ-strongly convex, i.e., [35]

$$\|\nabla F_i(\omega_1) - \nabla F_i(\omega_2)\| \leq L\|\omega_1 - \omega_2\|, \quad \forall \omega_1, \omega_2, \tag{20}$$

and

$$F_i(\omega_2) \geq F_i(\omega_1) + \nabla F_i(\omega_1)^\top(\omega_2 - \omega_1) + \frac{\mu}{2}\|\omega_2 - \omega_1\|^2. \tag{21}$$

Assume that the stochastic gradients are unbiased, i.e., $\mathbb{E}[\nabla F_i(\omega, \xi)] = \nabla F_i(\omega)$, and have bounded variance $\mathbb{E}[\|\nabla F_i(\omega, \xi) - \nabla F_i(\omega)\|^2] \leq \sigma^2$, as established in [36], [37]. In QAT, the quantisation process introduces additive noise simulated during local updates:

$$\tilde{\omega}_i^t = \omega_i^t + \epsilon_q^t, \quad \epsilon_q^t \sim \mathcal{U}(-S/2, S/2), \tag{22}$$

where $S = \frac{\omega_{\max} - \omega_{\min}}{2^b - 1}$ is the quantisation step size for bit-width b [38]. Using a first-order Taylor expansion, the local gradient under quantisation is approximated as:

$$\nabla F_i(\tilde{\omega}_i^t) \approx \nabla F_i(\omega_i^t) + H_i^t \epsilon_q^t, \tag{23}$$

where $H_i^t = \nabla^2 F_i(\omega_i^t)$ is the local Hessian matrix. Since ϵ_q^t is zero-mean and independent of ω_i^t, the expected gradient remains unbiased:

$$\mathbb{E}[\nabla F_i(\tilde{\omega}_i^t)] = \nabla F_i(\omega_i^t). \tag{24}$$

However, the variance of the stochastic gradient increases due to quantisation noise:

$$\mathbb{V}[\nabla F_i(\tilde{\omega}_i^t)] = \sigma^2 + \mathbb{E}[\|H_i^t \epsilon_q^t\|^2] \leq \sigma^2 + S^2\|H_i^t\|_F^2, \tag{25}$$

where $\|H_i^t\|_F$ is the Frobenius norm of the Hessian. Using standard analysis from [37], [39], the convergence bound of Federated SGD with simulated quantisation becomes:

$$\mathbb{E}[F(\omega_T)] - F(\omega^*) \leq \frac{L\|\omega_0 - \omega^*\|^2}{2\eta T} + \frac{\eta}{2}\left(\sigma^2 + S^2\|\bar{H}\|_F^2\right), \tag{26}$$

where ω^* is the optimal model and $\|\bar{H}\|_F^2$ is the average squared Frobenius norm across clients. The first term captures the $\mathcal{O}(1/T)$ convergence rate, while the second reflects the residual error floor. Notably, lower bit-widths b (e.g., 8-bit) increase the step size S, amplifying the error floor $S^2\|\bar{H}\|_F^2$. QAT mitigates this effect by embedding quantisation robustness into the training process [38], [40]. Unlike post-training quantisation (Post-Quant), QAT ensures that the model maintains performance even at low precision by adapting gradients to quantisation-induced distortions.

E. Global Aggregation Using FedDist

The FedDist algorithm improves the conventional aggregation process in FL by introducing a divergence analysis between quantised local and global model parameters. This selective aggregation strategy optimises model performance across highly diverse client data. Given quantised weights from the clients $\hat{\omega}_i^{t+1}$ and the global model $\hat{\omega}^t$, the Euclidean distance for divergence assessment is calculated as follows:

$$dist(\hat{N}_1, \hat{N}_2) = \sqrt{\sum_{i=1}^{C} (\hat{N}_1^{w_i} - \hat{N}_2^{w_i})^2}, \tag{27}$$

where $\hat{N}_1$ and $\hat{N}_2$ represent neurons in the local and global models, respectively, and C is the number of weights per layer. This is a layer-wise update process where the architecture of the global model is modified based on the diverging neuron. If the distance between the neurons of the local and global model is greater than the given threshold, the local node is considered as diverging; thus, layer-wise training is started. In this paper, the divergence threshold is defined as $\tau = 3\mu + \sigma$, where μ and σ represent the mean and standard deviation of neuron distance values, respectively. The values of μ and σ are computed dynamically within each global communication round across all neurons and participating clients, allowing the threshold to adapt in real time to the observed heterogeneity. Layers or neurons with distance values exceeding τ are considered divergent and excluded from aggregation in that round, improving stability under non-IID data distributions. The choice of the coefficient 3 follows the empirical 3σ rule, which statistically isolates extreme outliers (approximately 0.3% of values) while retaining normal update variations. This provides a stable, data-adaptive, and tuning-free boundary for divergence detection. Once there is no diverging neuron, the system performs the simple weighted average as the FedAVG algorithm. This technique enables the identification of the client, who constantly acts as an outlier and has a negative impact on the performance of the global model. The QAT FedDist algorithm is given in Algorithm 1.

V. Simulation Setup

This section explains the simulation setup created to thoroughly test the effectiveness of the proposed FFQ framework for HAR using CSI. Our goal is to validate the accuracy, robustness, and efficiency of the FFQ model under various conditions using experiments that mimic real-world scenarios in smart healthcare applications. In addition to the dataset description, we will discuss the performance metrics and also outline our performance evaluation strategy, which includes accuracy, communication overhead, and comparative analysis with state-of-the-art models.

A. Dataset Description and Data Partitioning

The raw dataset is collected in an indoor lab environment where an IEEE 802.11a standard is implemented using SDR. Two USRPs are used as transmitters and receivers and are kept approximately 6 m apart. The data is collected in a single physical environment with fixed transmitter-receiver positions, which ensures controlled signal integrity but limits environmental variation. Thirty volunteers from different age groups have participated in the data collection process. The participants also represent diverse physical characteristics, including gender, height, and body composition, introducing natural variability in movement patterns. The implementation and data collection process details are beyond the scope of this paper; however, the details on experimental setup can be seen in [20]. This dataset includes 80,000 samples across six typical daily activities: sitting, standing, running, falling, lying down, and jumping. Additionally, the line-of-sight component of CSI is also captured where no activity is performed. These activities were selected to encompass a range of human motions relevant to smart healthcare

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Algorithm 1: QAT-Enhanced FedDist Algorithm.

Input: Total communication rounds T, number of clients N, number of layers L
Output: Final global model $\hat{\omega}^T$

Server Execution:
Initialise global quantised model parameters $\hat{\omega}^0$
for $t = 0$ **to** $T - 1$ **do**
  // Client Updates
  for *client* $i = 1$ *to* N ***in parallel*** **do**
    Send $\hat{\omega}^t$ to client i
    $\hat{\omega}_i^{t+1} \leftarrow \text{ClientUpdate}(i, \hat{\omega}^t)$
  // Global Aggregation
  $\hat{\omega}^{t+1} \leftarrow \sum_{i=1}^{N} \frac{|D_i|}{\sum_{j=1}^{N} |D_j|} \hat{\omega}_i^{t+1}$ // Eq. 4
  // Euclidean Distance
  for $l = 1$ **to** $L - 1$ **do**
    for $i = 1$ **to** N **do**
      pair distance $\text{dist}(\hat{N}_1, \hat{N}_2)$ // Eq. 27
    Compute μ_l^t and σ_l^t for distances in layer l
    newNeuron $\leftarrow$ False
    for $c = 1$ **to** C // C: neurons in layer l **do**
      dist_threshold $\tau \leftarrow 3 \cdot \mu_{dl}^t + \sigma_{dl}^t$
      if $\text{mean}(c) > \tau$ **then**
        Append new neuron $\hat{\omega}_t^l$
        newNeuron $\leftarrow$ True
    if *newNeuron* **then**
      for $i = 1$ **to** N **do**
        $\hat{\omega}_i^l \leftarrow \hat{\omega}^l$ // Freeze layer l and below
        Update layers $\hat{\omega}_i^{l+1} \ldots \hat{\omega}_i^L$ at client i
  $\hat{\omega}^{t+1} \leftarrow \hat{\omega}^t$ // Global updates
Function ClientUpdate$(i, \hat{\omega})$:
  Initialise local model $\hat{\omega}_i \leftarrow \hat{\omega}$
  for *epoch* $= 1$ **to** E **do**
    foreach *batch* b *of size* B *from client* i*'s data* **do**
      $\hat{\omega}_i \leftarrow \hat{\omega}_i - \eta \nabla F_i(\omega_{i,\text{sim}}^t; b)$ // Eq. 18
  return $\hat{\omega}_i$

monitoring systems. The raw data comprises complex CSI for 52 subcarriers, capturing both amplitude and phase components. However, for this analysis, only the amplitude information is used to extract features, as discussed in Section IV.

To simulate a realistic FL environment, the dataset was equitably distributed among five clients (Edge Nodes - ENs). Each client was allocated 16,000 samples. Since the experimental setup involved only five edge clients in a cross-silo configuration, all clients actively participated in every global communication round during training. This full participation strategy ensures stable convergence and avoids the stochasticity that may arise from random client selection in large-scale cross-device FL settings.

A standard split of 80% training and 20% testing was employed to prepare the data for the federated training process. Hence, the training set of each EN consists of 12,800 samples. Notably, the test sets from each client were combined to form a comprehensive global test set, where the total number of test samples in the global test set is 16,000. This global test set plays a crucial role in assessing the overall performance of the proposed FFQ model, ensuring that the evaluation reflects a wide array of scenarios and is not biased by any single client's data profile. Additionally, the data distribution among the participants was kept highly unbalanced and statistically heterogeneous to reflect realistic non-IID FL scenarios. While this dataset captures significant user-level variability, we acknowledge its limitation in terms of spatial diversity. As discussed in Section VII, future work will extend this evaluation to cross-environment and multi-room settings to explore the generalisability of FFQ under diverse physical conditions.

B. Model Configurations and Hyperparameters

In this study, we employed a straightforward yet robust Deep Neural Network (DNN) for local learning. This model configuration is specifically chosen to balance the computational load and maintain high accuracy across various EN implementations. The local model consists of an input layer that accepts an N-dimensional vector with N as the number of features per sample. This architecture includes three fully connected layers with 128, 64, and 32 units, respectively, each followed by a rectified linear unit (ReLU) activation function. Finally, there is a dropout layer with a dropout rate of 0.3, and an output layer with a softmax activation function. Additionally, a regularisation factor of 0.01 was applied with a learning rate of 10^{-5}. The above hyperparameters were optimised through an extensive grid-search procedure conducted on a validation subset of the training data. The search explored learning rates in $\{10^{-4}, 10^{-5}, 10^{-6}\}$, dropout rates in $\{0.2, 0.3, 0.4\}$, and L_2 regularisation factors in $\{0.001, 0.01, 0.1\}$. The configuration with a learning rate of 10^{-5}, dropout 0.3, and regularisation 0.01 consistently achieved the best validation accuracy and stable convergence across all clients. Higher learning rates (10^{-4}) caused oscillations, whereas lower rates (10^{-6}) slowed convergence. Likewise, varying the dropout between 0.2 and 0.4 changed overall accuracy. These optimised values thus provide a balanced trade-off between convergence stability, communication efficiency, and generalisation.

C. Performance Metrics

To evaluate the proposed FFQ framework, we use average accuracy as the baseline metric. However, accuracy alone can be misleading with imbalanced datasets, leading to the accuracy paradox, where a model appears effective but is biased towards the majority class. To provide a more comprehensive evaluation, we also consider precision (Precision $= \frac{\text{TP}}{\text{TP + FP}}$), recall (Recall $= \frac{\text{TP}}{\text{TP+FN}}$), and the F1-score (F1-score $= 2 \times$

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$\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$). Precision measures the correctness of positive predictions, which is crucial for high-stakes activities like fall detection. Recall assesses the model's ability to identify all relevant instances, which is vital in scenarios where missing a positive instance is critical. The F1-score provides a harmonic mean of precision and recall, balancing both metrics and being especially useful for imbalanced classes. Here, TP is true positive, FP is false positive, and FN is false negative.

To evaluate the FFQ framework, we conducted experiments in a controlled simulation environment using a global test set, assessing the model's performance across all participants. In addition to classification accuracy, energy efficiency is considered a critical metric in FL, particularly in scenarios involving resource-constrained edge devices. We extend our energy model to capture both the computation and communication overhead involved in each communication round. The estimated energy consumption per round, E_{est}, is expressed as:

$$E_{\text{est}} = N \cdot (\gamma \cdot \text{FLOPs} + \beta \cdot P_{trn}), \tag{28}$$

where N is the number of participating clients, γ represents the energy cost per floating point operation (FLOP), which varies with bit precision, β is the communication energy cost per kilobyte, and P_{trn} is the model payload size per communication round in kilobytes. This generalised formulation supports fair evaluation across different training strategies, including full-precision and QAT. It reflects the trade-offs between computational workload and communication volume inherent in edge deployment scenarios. Furthermore, to ensure dimensional consistency of (28), we analyse the units:

$$\text{Units of } E_{\text{est}} = N \cdot \left(\frac{\text{J}}{\text{FLOP}} \cdot \text{FLOP} + \frac{\text{J}}{\text{KB}} \cdot \text{KB} \right) = \text{Joules}$$

The resulting unit of joules confirms the physical validity of the estimation.

VI. Results and Discussion

This section evaluates the proposed FFQ framework for HAR. The dataset, divided among 5 ENs, has 16,000 samples per node (12,800 for training and 3,200 for testing). For robust analysis, two distinct case studies are conducted, which include:

1) *Combined Temporal Fusion (CTF):* This setup explores the performance improvement achieved by integrating statistical and differential features with basic raw data features. The evaluation is conducted across two scenarios: multi-class classification of daily activities and binary-class fall detection.
2) *CTF with QAT:* In addition to the basic CTF setup, this case study incorporates QAT to investigate the impact on communication overhead and examine the trade-off between accuracy and energy efficiency. The analysis includes both multi-class and binary-class problems, comparing QAT and post-quantisation (where parameters are quantised after local training).

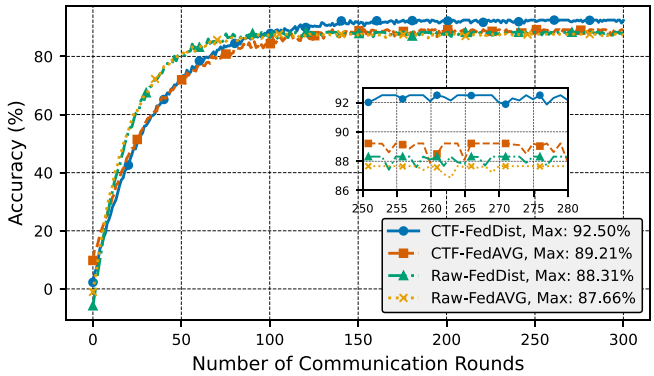


Fig. 3. The comparative accuracy learning curve for CTF and simple feature for FedDist and FedAVG algorithm.

A. Case:1 (a) CTF for Multi-Class Classification

The CTF aims to enhance classification accuracy by integrating statistical and differential features with raw data. We compare the FedDist and FedAVG algorithms with and without feature fusion in the multi-class classification of seven activities. The results in Fig. 3 present the learning curves from the global model evaluated using the test set across 300 communication rounds, comparing the accuracy of FedDist and FedAVG algorithms under two feature configurations: CTF and raw samples obtained using DWT and LDA. The CTF-FedDist setup achieved the highest accuracy, peaking at 92.50%, demonstrating the effectiveness of combining temporal features with the FedDist algorithm. In comparison, CTF-FedAVG, Raw-FedDist, and Raw-FedAVG reached maximum accuracies of 89.21%, 88.31%, and 87.66%, respectively. These results underscore the significance of incorporating contextual information and strategic model aggregation to enhance the accuracy of multi-class HAR.

The results in Fig. 4 show a set of normalised confusion matrices that represent the classification outcomes. Each confusion matrix displays the predictive performance across six activity classes, along with a No Activity category. The CTF-FedDist matrix shown in Fig. 4(a) indicates high diagonal values, which imply strong true positive rates for most activities, particularly for Fall, Sit, and No Activity. These results highlight the effectiveness of temporal features and strategic model aggregation. In comparison, the CTF-FedAVG matrix given by Fig. 4(b) also demonstrates high accuracy but slightly lower true positive rates for Run and Walk, indicating a minor increase in confusion between these two activities.

Conversely, the Raw-FedDist Fig. 4(c) and Raw-FedAVG Fig. 4(d) matrices exhibit lower diagonal values across activities, with a notable decrease in correctly classified instances for Fall, a critical class for safety in healthcare scenarios. Additionally, the off-diagonal elements represent misclassifications, with the non-zero values indicating the proportion of each actual class being incorrectly predicted as each of the other classes. The results also demonstrate that the Fall and Laydown are frequently confused due to the similar motion patterns captured using CSI. Similarly, activities like Walk and Run exhibit a degree of confusion, with off-diagonal values of 0.03 and 0.04, respectively.

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True/Pred	Fall	Lay	Sit	Stand	Run	Walk	NA
Fall	0.92	0.04	0.01	0.00	0.00	0.01	0.02
Laydown	0.03	0.91	0.01	0.01	0.01	0.00	0.02
Sit	0.00	0.00	0.97	0.01	0.00	0.01	0.00
Stand	0.01	0.01	0.00	0.96	0.01	0.01	0.00
Run	0.01	0.01	0.01	0.00	0.93	0.04	0.00
Walk	0.01	0.00	0.01	0.03	0.03	0.89	0.03
NA	0.00	0.01	0.02	0.02	0.01	0.02	0.91

(a) CTF-FedDist

True/Pred	Fall	Lay	Sit	Stand	Run	Walk	NA
Fall	0.88	0.05	0.03	0.01	0.00	0.01	0.02
Laydown	0.04	0.90	0.01	0.01	0.01	0.00	0.02
Sit	0.00	0.02	0.91	0.03	0.01	0.01	0.00
Stand	0.00	0.01	0.02	0.94	0.01	0.02	0.01
Run	0.01	0.01	0.02	0.01	0.87	0.06	0.02
Walk	0.01	0.00	0.01	0.03	0.04	0.88	0.03
NA	0.01	0.01	0.01	0.02	0.01	0.02	0.90

(b) CTF-FedAVG

True/Pred	Fall	Lay	Sit	Stand	Run	Walk	NA
Fall	0.87	0.06	0.03	0.01	0.00	0.01	0.02
Laydown	0.04	0.90	0.01	0.01	0.01	0.00	0.02
Sit	0.00	0.02	0.91	0.03	0.01	0.01	0.02
Stand	0.00	0.01	0.02	0.92	0.02	0.03	0.01
Run	0.01	0.01	0.02	0.03	0.82	0.08	0.02
Walk	0.01	0.00	0.01	0.03	0.05	0.87	0.03
NA	0.01	0.02	0.01	0.03	0.02	0.02	0.88

(c) Raw-FedDist

True/Pred	Fall	Lay	Sit	Stand	Run	Walk	NA
Fall	0.85	0.06	0.03	0.02	0.01	0.01	0.02
Laydown	0.06	0.88	0.02	0.01	0.01	0.00	0.02
Sit	0.00	0.02	0.87	0.05	0.01	0.01	0.02
Stand	0.01	0.01	0.01	0.94	0.02	0.01	0.00
Run	0.02	0.00	0.03	0.02	0.84	0.06	0.02
Walk	0.02	0.00	0.01	0.02	0.07	0.83	0.03
NA	0.01	0.02	0.02	0.02	0.02	0.04	0.87

(d) Raw-FedAVG

Fig. 4. Comparison of FedDist and FedAVG performance using CTF and amplitude-only features for multi-class HAR. NA denotes no activity.

TABLE II
PERFORMANCE COMPARISON FOR CTF-FEDDIST AND RAW-FEDDIST

	CTF-FedDist			Raw-FedDist		
Class	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Fall	0.939	0.920	0.929	0.926	0.870	0.897
Laydown	0.929	0.919	0.924	0.882	0.909	0.896
Sit	0.942	0.980	0.960	0.901	0.910	0.905
Stand	0.932	0.960	0.946	0.868	0.911	0.889
Run	0.939	0.930	0.935	0.882	0.828	0.854
Walk	0.908	0.890	0.899	0.853	0.870	0.861
NA	0.929	0.919	0.924	0.880	0.889	0.884

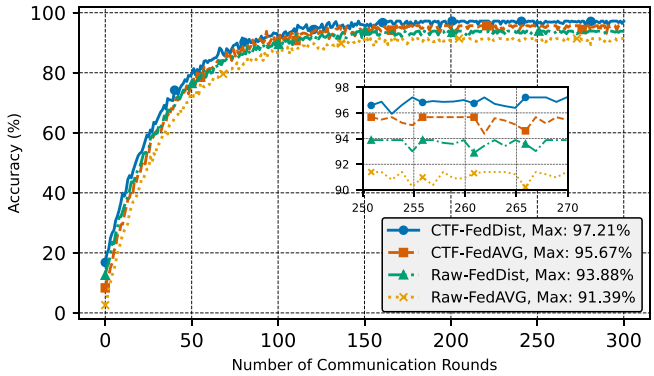


Fig. 5. The comparative accuracy learning curve for CTF and simple feature for FedDist and FedAVG algorithm for fall detection.

In Table II, there is a comparison of performance metrics that shows the benefits of using feature fusion instead of simple raw features in the FedDist algorithm. The CTF-FedDist configuration has a higher precision, recall, and F1 Score when compared to Raw-FedDist, especially for critical activity classes such as Fall, Run, and Stand. It is important to note that in the Fall classification, the precision and F1 Score of CTF-FedDist is significantly higher than Raw-FedDist.

B. Case:1 (b) CTF for Fall Detection (Binary Classification)

In this scenario, a binary classifier is trained to identify Falls from other daily routine activities. The model is trained for 300 communication rounds, and a global test set is used to evaluate its performance. Results for the accuracy learning curve for binary classifiers with and without temporal feature fusion are given in Fig. 5. These curves exhibit classical behaviour, with a sharp increase in initial results and a steadier behaviour state after 100 communication rounds. The results show that the CTF fusion improves the model's performance. Notably, the CTF-FedDist configuration achieves the highest accuracy, peaking at 97.21%, illustrating the benefits of integrating temporal context in the detection algorithm. Moreover, this performance is closely followed by the CTF-FedAVG, with a maximum accuracy of 95.67%. On the other hand, configurations that used raw features such as Raw-FedDist and Raw-FedAVG exhibited reduced peak accuracies of 93.88% and 91.39%, respectively.

The results in Fig. 6 show the confusion matrices for binary class Fall detection using CTF and raw features for FedDist and FedAVG algorithms. The matrices reveal that CTF features combined with FedDist, as shown in Fig. 6(a), yield the highest true positive rate for Falls at 0.98% and a true negative rate of 0.96%, outperforming the other configurations. In comparison, the raw feature sets given in Fig. 6(c), (d) exhibit reduced performance, particularly in distinguishing non-fall activities.

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True / Pred	Fall	Non-Fall
Fall	0.98	0.02
Non-Fall	0.04	0.96

(a) CTF–FedDist

True / Pred	Fall	Non-Fall
Fall	0.96	0.04
Non-Fall	0.06	0.94

(b) CTF–FedAVG

True / Pred	Fall	Non-Fall
Fall	0.95	0.05
Non-Fall	0.08	0.92

(c) Raw–FedDist

True / Pred	Fall	Non-Fall
Fall	0.93	0.07
Non-Fall	0.10	0.90

(d) Raw–FedAVG

Fig. 6. Confusion matrices for binary fall detection using CTF and amplitude-only features with FedDist and FedAVG.

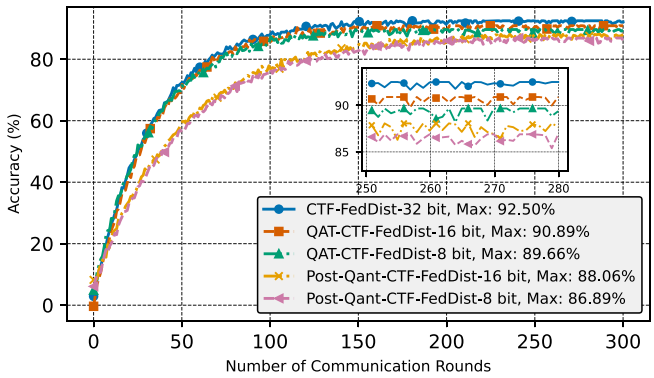


Fig. 7. The comparative accuracy learning curve for Post-Quant and QAT for Multi-class classification using FedDist.

C. Case:2 CTF With QAT

Building upon the insights from the previous two case studies, we now turned our attention to model compression via quantisation at varying bit precisions and incorporating QAT. This case study is driven by optimising the FFQ framework, particularly the FedDist algorithm, to balance performance (accuracy) and reduce communication overhead. Furthermore, this case study evaluates the impact of lowering model parameter precision from the standard 32-bit to 16-bit and further to 8-bit, both with QAT and post-quantisation.

Fig. 7 illustrates the multi-class classification accuracy over 300 communication rounds, showing the convergence behaviour of different precision levels. The 32-bit CTF-FedDist baseline achieved the accuracy of 92.50%. With quantisation, QAT-FedDist at 16 and 8-bit achieved 90.89% and 89.66%, respectively, confirming the effectiveness of QAT in mitigating accuracy loss under reduced precision. In comparison, Post-Quant models reached 88.06% (16-bit) and 86.89% (8-bit), underscoring the importance of QAT in optimising the model for lower-bit precision.

Fig. 8 presents the learning accuracy for binary Fall detection, illustrating the performance for models trained with and without QAT. Benchmarking against the CTF-FedDist 32-bit precision model, which achieved an accuracy of 97.21%, the QAT-FedDist models at 16 and 8-bit precisions achieve comparable accuracies

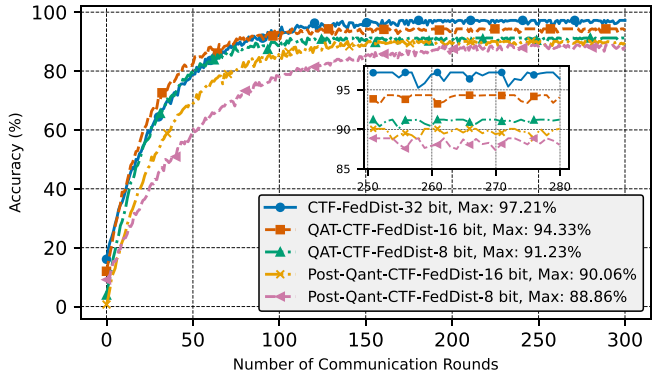


Fig. 8. The comparative accuracy learning curve for Post-Quant and QAT for binary fall detection using FedDist.

of 94.33% and 91.23%, respectively. In contrast, the Post-Quant with FedDist at 16 and 8-bit quantisation demonstrated a lower average accuracy of 90.06% and 88.86%. Although a certain degree of accuracy reduction is anticipated due to model compression, applying QAT considerably mitigates this effect, showcasing its role in maintaining high model performance despite the reduced bit depth.

D. Energy Efficiency Evaluation Under QAT

The energy efficiency of the proposed FFQ framework was evaluated using (28), which accounts for both computation and communication energy. For communication, an energy coefficient of $\beta = 0.3$ mJ/KB was used, representing typical 5G transmission cost [41]. The payload sizes P_{trn} for 32, 16, and 8-bit precisions were 185.91 KB, 98.33 KB, and 52.31 KB, resulting in respective communication costs of 55.773 mJ, 29.499 mJ, and 15.693 mJ per round. For computation, the model (input: 52, output: 7) required ≈68,480 FLOPs per round; based on [42], this corresponds to 0.314 mJ (32-bit), 0.0068 mJ (16-bit), and 0.0027 mJ (8-bit).

While computation costs are relatively negligible compared to communication, their inclusion offers a complete view of the energy profile. The total energy per client per round was 56.087 mJ (32-bit), 29.506 mJ (16-bit), and 15.696 mJ (8-bit). Quantisation therefore improves energy efficiency by **47.4%** and **72.0%** at 16- and 8-bit precisions, respectively, relative to the 32-bit baseline. This highlights the effectiveness of the FFQ framework in energy-constrained FL environments. A comparative summary of the estimated energy consumption under each quantisation configuration is shown in Fig. 9, where the 8-bit QAT setting achieves the lowest energy footprint while maintaining acceptable model performance.

E. Quantisation Trade-Off and Scalability Analysis

Quantisation reduces model size and communication overhead in FL but introduces discretisation noise from reduced precision. We adopt QAT, which simulates quantisation during training to preserve accuracy. As shown in Figs. 7 and 8, QAT-FedDist limits accuracy loss to 1.61% (16-bit) and 3.84% (8-bit) for multi-class HAR, and 2.88% and 5.98% for binary fall

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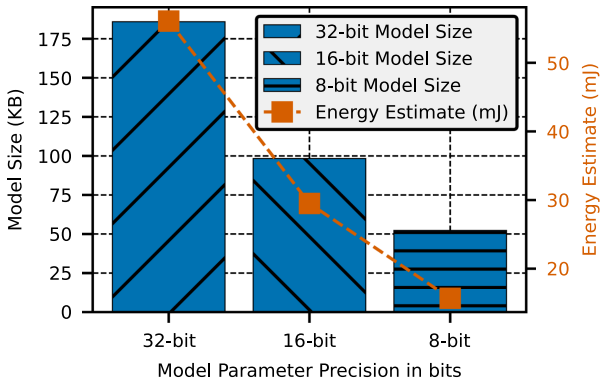


Fig. 9. Energy estimate for one EN per communication round, combining computation and communication costs under different quantisation settings.

detection, consistently outperforming post-training quantisation at the same bit-widths.

Theoretically, QAT injects zero-mean quantisation noise; under L-smooth and μ-strongly convex objectives, it adds a bounded error floor $\mathcal{O}(S^2\|H\|_F^2)$ to the convergence in (26), encouraging flatter minima that are robust to low-precision perturbations. This aligns with the analysis in Section IV-D4, where the bound includes $\frac{\eta}{2}(\sigma^2 + S^2\|\tilde{H}\|_F^2)$. Empirically, the small degradations even at 8-bit indicate our training reduces sensitivity to the quantisation step size S, yielding stable convergence under reduced precision. Thus, FFQ+QAT provides communication/energy savings while maintaining robust convergence across precision levels and edge conditions.

Additionally, our experiments simulate a realistic federated setup with five heterogeneous clients, we acknowledge that real-world deployments often involve significantly larger and more diverse edge networks. The FFQ framework supports scalability through two primary mechanisms: (i) model compression via QAT, which reduces communication overhead by up to 71%, and (ii) the FedDist aggregation strategy, which improves convergence stability across non-IID clients. These features make FFQ suitable for resource-constrained devices and high-client-volume settings.

VII. Conclusion and Future Work

This paper presented the FFQ framework for indoor HAR using CSI, integrating advanced pre-processing, feature fusion, and model compression. The proposed FFQ framework adopted the FedDist algorithm, which is tailored to mitigate overfitting by adjusting model parameters based on neuron dissimilarity. Our evaluation shows that CTF significantly improves classification accuracy, with a 4.29% increase in multi-class HAR and a 5.55% boost in binary fall detection compared to models with raw feature fusion. Additionally, model compression with QAT achieved a 16-bit quantisation, reducing communication overhead by 47% with minimal accuracy loss (1.61% for multi-class and 2.88% for fall detection). The 8-bit quantisation further reduced overhead by 71%, with performance decrements of 3.84% and 5.98%, respectively. Future work will focus on analysing computational complexity and evaluating scalability across diverse datasets and multi-client federated environments where resource heterogeneity may affect performance. Although current experiments used CSI data from 30 participants in a controlled indoor setup, future validation will include cross-environment deployment in smart homes, healthcare facilities, and multi-room testbeds to assess robustness under varied propagation conditions.

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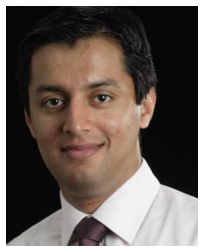
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